ELSEVIER

Tectonophysics 244 (1995) 251–266

TECTONOPHYSICS

Palaeomagnetic study of block rotations in the Niksar overlap region of the North Anatolian Fault Zone, central Turkey

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Received 16 March 1994; revised version accepted 18 October 1994

Abstract

This palaeomagnetic study investigates crustal deformation within, and adjacent to, the Niksar overlap area of the North Anatolian Fault Zone (NAFZ) in central-east Turkey. The studied rock formations comprise: (1) red limestones of Late Cretaceous age (3 sites); (2) mafic lavas of Eocene age on the north side (13 sites) and south side (9 sites) of the NAFZ; and (3) volcanic rocks of Pliocene–Quaternary age from the Niksar pull-apart basin within the NAFZ (8 sites). Comparisons with reference palaeofield directions computed from apparent polar wander paths of the Eurasian and Afro–Arabian plates identify two scales of regional and local tectonic rotation:

(1) A pre-tilting remanence in the Eocene volcanic rocks south of the NAFZ ($D/I = 144.1/-47.5°$, $\alpha_{95} = 7.6°$) is interpreted to reflect counterclockwise rotation by 30–40° from the reference palaeofields. Contemporaneous volcanic rocks from the north side of the NAFZ have the same reverse polarity recorded in pre-tilting magnetisations. The remanence is also rotated counterclockwise ($D/I = 152.4/-42.5°$, $\alpha_{95} = 11.3°$), but by about 8° less than the volcanics on the south side of the NAFZ. Hence similar amounts of rotation are observed on both sides of the NAFZ and are interpreted to reflect motions during the pre-Middle Miocene collisional history in this sector of the Pontides. No distributed clockwise rotations anticipated from subsequent dextral motion along the NAFZ intracontinental transform are observed. The slightly larger anticlockwise rotation found on the south side of the NAFZ probably records relative rotation of en-echelon wedges by continental escape during post-Middle Miocene strike slip along the transform.

(2) Within the narrow zone of intense deformation along the NAFZ, Cretaceous limestones appear to be rotated clockwise by dextral strike-slip motion whilst Plio–Quaternary lavas within a fault-bounded block in the overlap region associated with the Niksar pull-apart basin, have magnetisations consistently directed 240–270°E. Magnetic inclinations are not diagnostic of polarity but both polarity solutions identify rapid clockwise rotation at rates in excess of 50°/m.y. A normal polarity solution is favoured and implies that a block (ca. 5 km across) has undergone a strike-slip displacement of around 12 km within the NAFZ during the last polarity chron. Cretaceous–Eocene palaeolatitudes are closer to those predicted from Eurasia than Afro–Arabia, but a study of older rocks is required to resolve affinities of this sector of the Anatolian block.

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SSDI 0040-1951(94)00241-X

Theoretical models of crustal deformation across intracontinental transforms obeying power-law behaviour and treating the lithosphere as a viscous medium predict that distributed clockwise rotations should be observed about as broad across the NAFZ. These rotations are not observed. Instead intense clockwise rotation is confined to a narrow zone close to the major fault break. The concentration of historic seismic activity here also implies that the bulk of the dextral motion between the Eurasian Plate and the Anatolian block is accommodated by slip along faults close and parallel to the main trace of the NAFZ.

1. Introduction

Palaeomagnetic studies in the vicinity of large intracontinental transform faults in the San Andreas Zone of western North America (e.g., Hornafius et al., 1986) and the Middle East (Ron et al., 1984) have identified large rotations over distributed zones about the main fault lineament. Wider study of these transforms is required to determine whether this is generally true and if such deformations can be accommodated by theoretical models of lithosphere behaviour (Sonder et al., 1986). The North Anatolian Fault Zone (NAFZ) is the intercontinental expression of a major dextral transform separating the Eurasian Plate from the Anatolian block and Afro–Arabian Plate. The present study is centred on the town of Niksar located at the western edge of a right-stepping overlap region in central Turkey (Fig. 1). Dextral motion across the NAFZ is concentrated mainly within a narrow zone incorporating pull-apart basins, but is laterally distributed over a wider area by a sequence of NE–SW-trending fault splays branching from the main system.

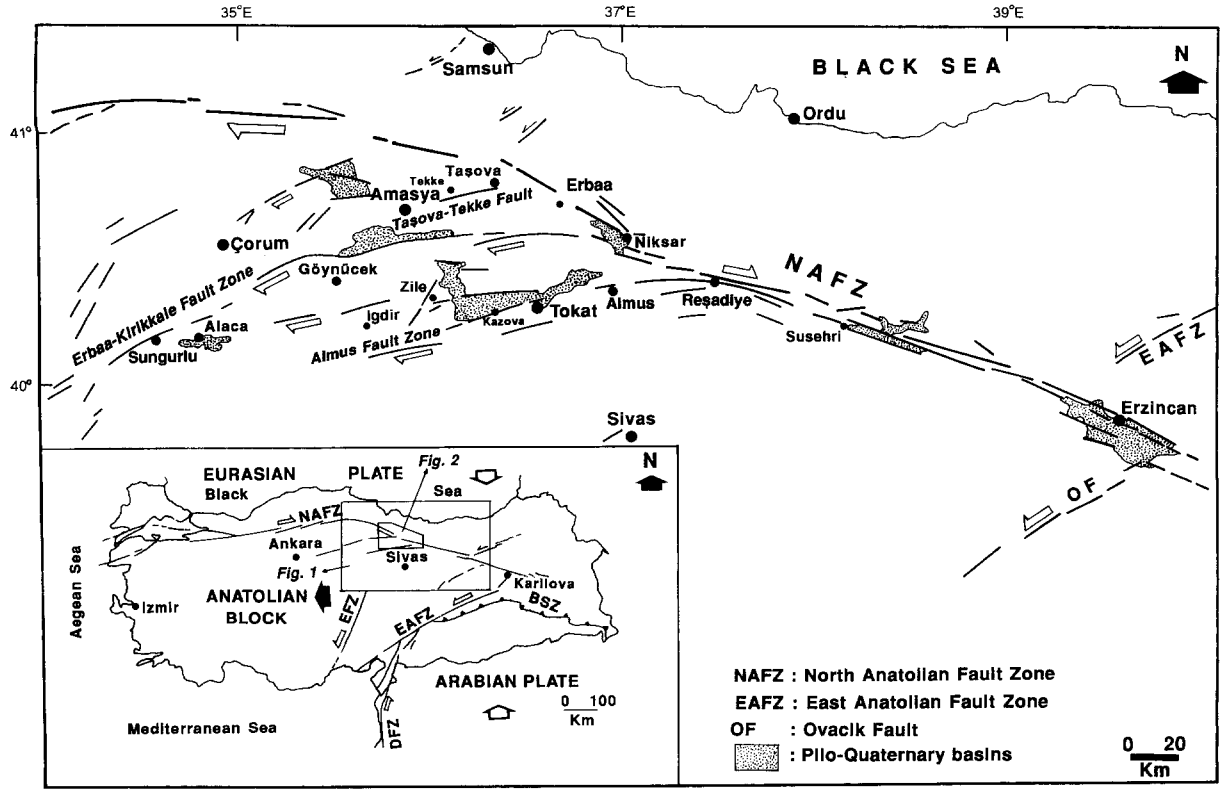


Fig. 1. Regional structural map illustrating the splay fault system branching from the NAFZ (simplified from the 1:2,000,000 Geological Map of Turkey).

These faults segment the crust into a set of large blocks (about 5–10 km across) believed to be undergoing rotations to accommodate the dextral motion across the plate boundary. The overlap region is about 8–13 km long and 12 km across and is cut by prominent NE–SW, NW–SE and E–W faults, all showing dextral displacement senses. The principal displacement direction of the main fault system here is WNW–ESE (Tatar, 1988, 1993; Temiz, 1989).

The 1942 earthquake fault bounds the northern margin of the Niksar pull-apart basin, while the 1939 earthquake fault bounds the southern margin (Fig. 1). The strike-slip fault geometry is very complex at the eastern margin of the basin where the main fault segments appear to consist of secondary Riedel shears. In a study of the neighbouring Reşadiye region, Seymen (1975) divided the fault segments into three sub-zones on the basis of their trends: (1) the Ladik–Niksar–Bereketli Fault Zone trending roughly E–W; (2) the Hanyeri–Gökçebel Fault Zone trending NE–SW; and (3) the Kelkit Valley Fault Zone trending NW–SE approximately parallel to the main fault zone. The eastern part of the Kelkit Valley Zone forms the southern border to this complex faulted area. Seymen (1975) suggested that secondary faults developed between these subzones and rotated in an anticlockwise sense.

Three main splay faults branch from the North Anatolian Fault Zone in the Niksar and Reşadiye regions (Fig. 1). These comprise: (1) the Almus Fault Zone (Bozkurt and Koçyiğit, 1992), 2) the Erbaa–Kırıkkale Fault Zone (corresponding with the 1939 earthquake fault break) (Ketin, 1969; Şengör et al., 1989); and (3) the Taşova–Tekke

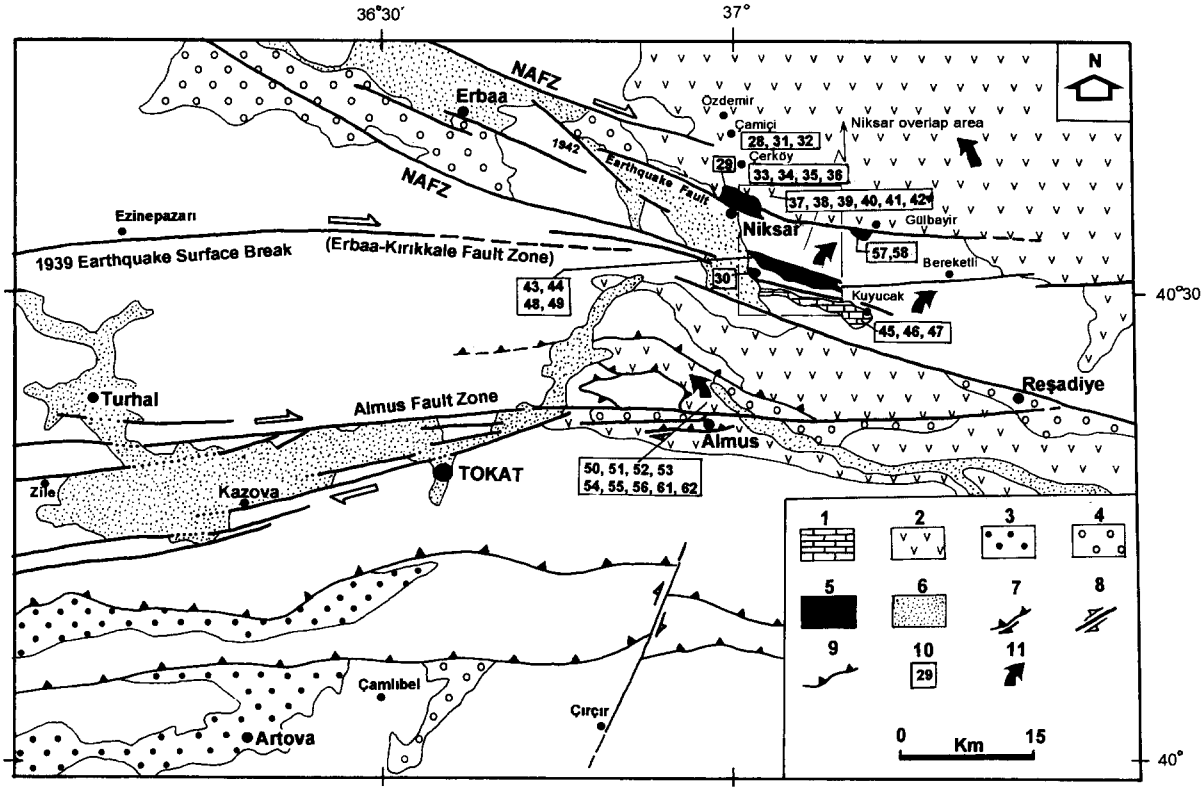


Fig. 2. Simplified geological map of the study area, showing sampling localities. The regional location is shown in Fig. 1. The inset legend: *1* = Upper Cretaceous red limestones; *2* = Eocene volcano-sedimentary units; *3* = Miocene–Pliocene deposits; *4* = Plio–Quaternary deposits; *5* = Plio–Quaternary volcanics; *6* = Quaternary deposits; *7* = transpressional fault with reverse component; *8* = strike-slip fault; *9* = thrust; *10* = sampling locations and numbers of site; *11* = sense of rotation.

Fault Zone. The Almus Fault Zone (Bozkurt, 1990; Koçyiğit and Bozkurt, 1991; Bozkurt and Koçyiğit, 1992) is an active dextral strike-slip fault system extending for about 150 km from Reşadiye in the east to Iğdir town, southeast of Zile, in the west. It has well developed surface expression with a width ranging from a few hundred metres along its eastern course to 12 km at its western end, where an active strike-slip depression, the Kazova basin, is located (Bozkurt and Koçyiğit, 1992). The Kırıkkale–Erbaa Fault Zone extends from the Erzincan basin to Sungurlu in the west, a length of approximately 350 km. It bounds the southern margin of the Niksar basin, where it changes direction from NW–SE to E–W, and was first recognised by Ketin (1957) following the Greater Erzincan earthquake in 1939. A third major splay fault (authors' observations) branches from the main fault zone in the southern part of Taşova and trends 70–80°E with a length of about 25–30 km to near Tekke. In addition to these main splays, several discontinuous lineaments are situated west of Tokat. Each splay fault bends at the end of its course to isolate several smaller blocks which are deforming in a complex way.

The present study uses palaeomagnetism to determine the magnitude and sense of rotation within the Niksar overlap region of this fault system. The technique is well suited to this area because lavas and tuffs, which can usually be accurately referred to the palaeohorizontal, are widely distributed, both temporally and spatially. Suitable magnetic facies comprising red limestones and volcanic rocks were identified by initial study and the subsequently cored sampling sites (Fig. 2) highlight the rocks of potential palaeomagnetic value. They can be divided into three main groups on the basis of their age and tectonic relationships and comprise: (a) Upper Cretaceous red limestones; (b) Eocene volcano-sedimentary deposits; and (c) Pliocene–Quaternary volcanic rocks.

2. Field and laboratory methods

The palaeomagnetic samples were collected initially as oriented blocks, and subsequently as cores drilled with a portable two-stroke motor and oriented in situ. Local tilts were recorded at outcrop. These can be ascertained most reliably in the sedimentary units; in the volcanics inter-basaltic horizons, ropy flow surfaces and vesicle flow were used to determine dip of the studied units. The Upper Cretaceous red limestones occur at three localities around Kuyucak village (sites 45–47). Eocene lavas were sampled at 13 localities (sites 28, 31–42) located within a large outcrop north of the Niksar basin and mainly to the north of the North Anatolian Fault Zone (NAFZ). Lavas of the same age comprise a succession in the Almus region on the opposite (south) side of the NAFZ and sites 50–56 and 61–62 are a complete section through a lava sequence here. Samples of Pliocene–Quaternary lavas were collected at eight localities sited within the overlap region around Niksar and west of Bereketli (29, 30, 43, 44, 48, 49, 57 and 58); these lavas are considered to be linked to the formation of the Niksar pull-apart basin (Tatar et al., 1992).

A preliminary study of oriented hand samples highlighted rock facies yielding stable magnetic component structures. A second collection was then cored using portable drills; between 5 and 9 separate cores were drilled at each location and the samples oriented by both sun and magnetic compasses whilst still attached to the outcrop. Site locations are given in Fig. 2. In the laboratory, the cores were sliced into 2.4 cm long cylinders for routine demagnetisation studies. The component structure of the remanence was evaluated in most samples by thermal demagnetisation in steps of 50 or 100°C to 500°C and then typically in 20°C steps to 560°C or 620°C to subtract components resident in magnetite. Two or three cores from each site in the volcanic rocks were stepwise demagnetised in alternating fields. The component structures of the natural remanent magnetisations (NRMs) were identified by study of orthogonal vector plots; the constituent components were then isolated interactively and their directions calculated by principal component analysis.

Typically, a range of two- and three-component behaviours is identified in these rocks as illustrated by representative examples (Figs. 3, 4

Table 1
Predicted reference palaeofield directions of normal polarity at the location of the study area in central Turkey

	Eurasia D/I	Afro–Arabia D/I
Recent	0/58	0/58
Early Tertiary	8/47	4/50
Late Cretaceous	17/46	346/33
Early L. Cretaceous	7/19	331/16

and 6–8). Lowest unblocking temperature components are evidently mostly of a viscous origin and are not discussed further; in most samples, discrete and convergent higher blocking temperature components are characteristic remanent magnetisations (ChRM's) and used to calculate the site means listed in Table 2. For identification of magnetic polarity and initial analysis these components may be compared with the present normal ($D/I = 0/+59.8°$) and reverse ($D/I = 180/-59.8°$) geomagnetic field directions in the study area. This comparison identifies net tectonic rotations about horizontal and vertical axes for Pliocene and younger rocks. Plate motions since Late Cretaceous times necessitate revision of these comparative field directions going backwards in time. Reference field directions may be calculated from the apparent polar wander paths from the Eurasian and Afro–Arabian blocks (Piper, 1987). The palaeofield migration paths calculated for this location indicate a field direction moving slightly to the west (Eurasia) or the east (Afro–Arabia) or north, going backwards in time (Table 1). The local field inclination is predicted to have increased progressively since Cretaceous times.

The age of remanence acquisition in deformed rocks is evaluated by applying the palaeomagnetic fold test. If the collection is not numerous enough, or deformation is insufficiently diverse, for this test to be effective, the remanence direction adjusted for tilt about the local strike may nevertheless yield an indication of the origin of the remanence: if the direction is closer to the palaeofield inclination in stratigraphic coordinates, the remanence is likely to predate tilting; if closer in geographic coordinates a secondary overprint postdating deformation is suspected.

3. Palaeomagnetic results

Site mean characteristic magnetisations (ChRM's) together with relevant statistics, are summarized in Table 2 and considered in order of rock age.

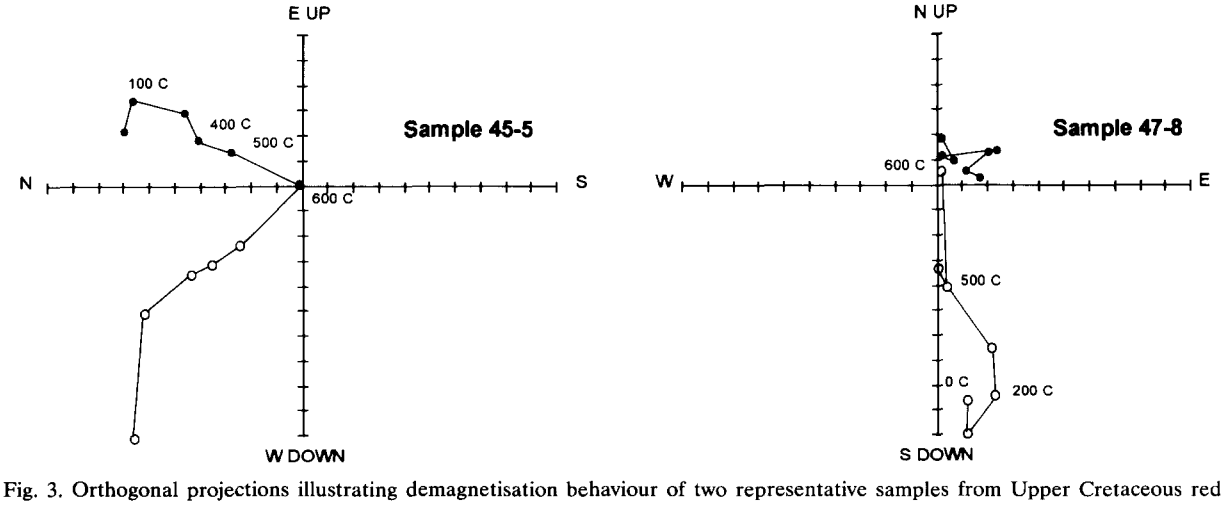


Fig. 3. Orthogonal projections illustrating demagnetisation behaviour of two representative samples from Upper Cretaceous red limestones. The orthogonal projections illustrate the behaviour of the remanence to progressive steps of thermal demagnetisation (°C) as projections onto the horizontal (dots) and vertical (circles) planes. The listed temperature steps are selected to define the trajectory shown. Total NRM intensities are 0.12 (45) and 0.04 (47) $\times 10^{-5}$ A m^{-2} kg^{-1}. All directions are shown in situ.

3.1. Upper Cretaceous red limestones

Thermal demagnetisation defines weak and somewhat erratic magnetisations (e.g., sample 47-8 in Fig. 3) in the Upper Cretaceous pelagic red limestones southeast of Niksar. Above 100–200°C these are essentially convergent and yield the site mean characteristic remanent magnetisations listed in Table 2. The directions have intermediate positive inclinations both in situ and in bedding coordinates; hence they are of normal polarity and may well have been magnetised during the Cretaceous normal superchron (118–83 m.y.). The ChRM's at sites 46 and 47 diverge widely when tilt-adjusted and these magnetisations are possibly post-folding. If secondary, the magnetisations are strongly rotated (at least 97–101° clockwise) with respect to the reference palaeofield directions (Table 2). Site 45 also appears to be rotated clockwise; in this case the tilt-adjusted direction is closer to the predicted field. These three sites are unlikely to be within the same tectonic block and in view of the small amount of data do not merit further interpretation.

3.2. Eocene volcanic rocks north of the NAFZ

An Eocene volcano-sedimentary sequence (formerly mapped as Cretaceous on the 1:500,000 geological map of Turkey; sheet Samsun, Göksu et al., 1974) comprises a large outcrop, but probably represents a relatively thin (ca. 200 m) succession immediately to the north and northeast of Niksar. More recent stratigraphic evidence given by Blumenthal (1950) and quoted in unpublished reports of the Geological Survey of Turkey (M.T.A.) assigns these rocks to Eocene times. The outcrop is cut by NW–SE-trending faults and lies mostly north of the main trace of the NAFZ. The palaeomagnetic samples come from road cuts at 13 localities around the villages of Özdemir, Çamiçi, Cerköy and Gülbayır; most of these outcrops are in layered basalt flows typically 2–6 m in thickness.

The majority of samples exhibit moderate to high degrees of stability. Convergent ChRM's are subtracted close to the Curie point of magnetite in most samples (Fig. 4); a few cores have small unresolved hematite components (e.g., sample

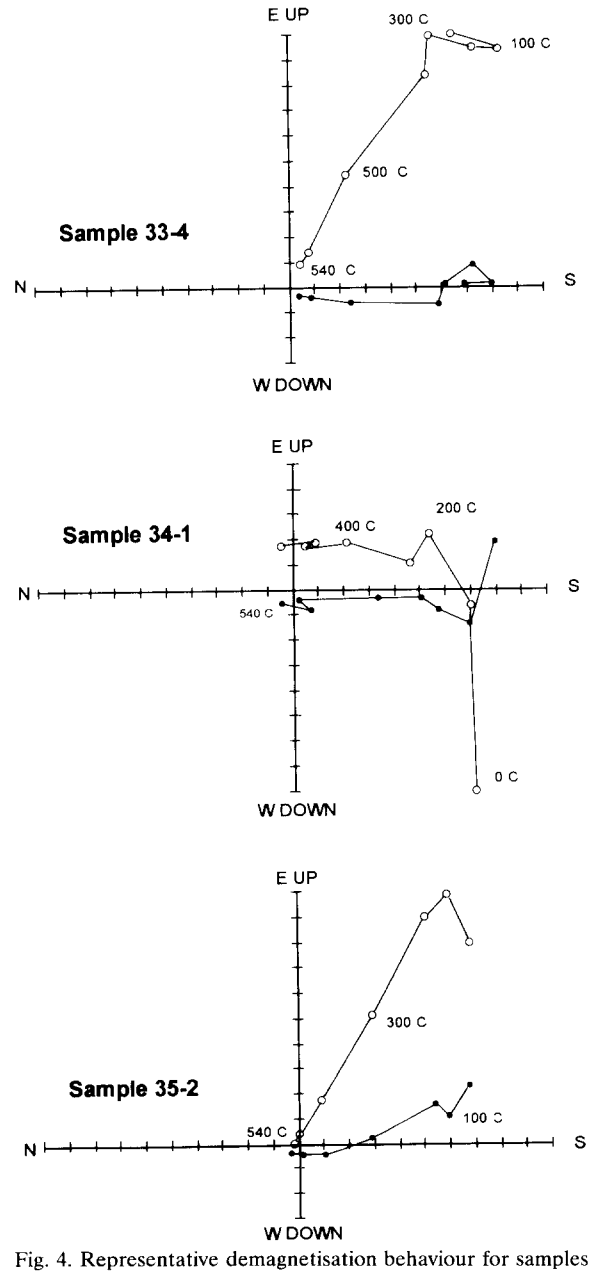


Fig. 4. Representative demagnetisation behaviour for samples from the basaltic lavas of Eocene age north of the NAFZ. Symbols are as for Fig. 3 and directions are in situ. Total NRM intensities are 94 (33), 35 (34) and 136 (35) $\times 10^{-5}$ A m^{-2} kg^{-1}.

Table 2
Palaeomagnetic results from Upper Cretaceous, Eocene and Pliocene–Quaternary rocks, Niksar region, central Turkey

Site	Age	N/n	R	α_{95}	k	D° (in situ)	I° (in situ)	D° (tilt adj.)	I° (tilt adj.)
(1) *Upper Cretaceous pelagic red limestones*:									
45	K	7/7	6.69	14	19.2	36.0	34.9	66.6	55.5
46		7/7	6.72	13	21.4	96.9	49.5	11.7	55.6
47		12/11	10.50	11	19.9	101.4	66.4	177.9	46.3
(2) *Eocene volcano-sedimentary succession north of NAFZ*									
33	Te	7/7	6.97	5	175.1	182.3	−54.4	175.4	−33.5
34		7/5	4.71	21	13.9	166.5	−3.9	158.0	−24.8
35		7/5	4.99	4	432.2	170.0	−56.7	171.3	−50.8
36		7/6	5.98	4	255.3	168.9	−67.0	173.5	−48.2
37		7/7	6.96	5	167.1	171.7	−26.7	158.3	−69.6
38		7/7	6.36	21	9.3	138.7	−7.9	127.9	−26.5
39		7/6	5.55	21	11.0	140.4	−16.4	152.9	−38.8
40		4/3	2.96	18	50.6	132.6	7.0	136.8	−41.6
41		5/5	4.75	20	16.0	115.8	−14.0	128.9	−38.3
42		3/3	2.98	14	82.7	128.5	−16.8	151.2	−40.8
Mean, all sites in situ:									
		10	8.66	20.1	6.7	147.7	−26.9		
Mean, all sites tilt adjusted:									
		10	9.53	11.3	19.2			152.4	−42.5
(Palaeomagnetic pole: 280.8°E, 62.0°E, dp/dm = 8.6/13.9°)									
(3) *Eocene volcanic succession south of NAFZ, Almus*									
50	Te	8/8	7.96	4	177.3	106.3	−41.7	141.2	−40.4
51		8/7	6.85	10	39.4	134.4	−38.4	151.1	−57.3
52		9/9	8.81	8	42.2	127.2	−9.2	141.6	−47.2
53		10/10	9.84	6	57.0	121.7	−24.3	132.0	−49.0
54		8/8	7.94	5	124.0	128.0	−52.2	154.0	−55.2
55		8/6	5.77	15	22.1	104.5	−38.2	132.3	−54.7
56		9/9	8.89	6	70.6	127.4	−39.5	154.9	−44.4
61		7/6	5.82	13	27.7	136.0	−25.9	145.8	−28.9
62		6/5	4.92	11	51.5	28.4	56.9	38.1	39.8
Mean, all sites except 62, in situ:									
		8	7.72	11.4	24.7	123.5	−34.2		
Mean, all sites except 62, tilt adjusted:									
		8	7.87	7.6	54.0			144.1	−47.5
(Palaeomagnetic pole: 296.4°E, 58.4°N, dp/dm = 6.4/9.9°)									
(4) *Pliocene–Quaternary volcanic rocks, Niksar overlap region*									
29	Tp–Q	7/7	6.94	6	95.9	179.7	88.0	199.3	−21.9
30		7/7	6.93	7	83.1	261.0	−26.9	260.2	29.1
43		3/3	2.99	9	201.3	264.2	10.8	262.9	1.9
44		6/6	5.70	17	16.7	243.2	23.9	240.7	11.2
48		7/5	4.70	22	73.1	242.3	19.3	242.6	−16.1
49		7/6	5.93	8	72.8	255.1	71.8	235.7	59.4
57		8/8	7.97	4	205.0	0.3	67.3	15.5	23.0
58		10/10	9.84	7	57.0	303.8	57.7	244.7	32.2

D and I are the mean declination and inclination derived from n samples of a total site population of N measured cores. R is the resultant vector, α_{95} is the radius of the cone of 95 percent confidence about the mean direction, and k is the Fisher precision estimate [$= (n-1)/(n-R)$]; dp and dm are the radii of the oval of confidence about the pole position in the colatitude direction and at right angles to it, respectively. The reference mean palaeomagnetic direction in the Niksar district (36.8N, 40.6E) for Late Tertiary–Recent times is $D/I = 0°/+57.9°$ (normal) and $D/I = 180°/-57.9°$ (reversed).

34-1 in Fig. 4). Although most cores show individual stability at the northern sites (28, 31 and 32) on the lava plateau in the vicinity of Özdemir, no within-site groupings are recognised here; the origin of this varied remanence record is uncertain and requires rock magnetic investigations. The remaining sites all have reversely magnetised ChRM's with variable inclination

Following adjustment for tilts measured at outcrop the declinations group more closely in a south-southeast quadrant and all sites have intermediate negative inclinations (Fig. 5). The im-

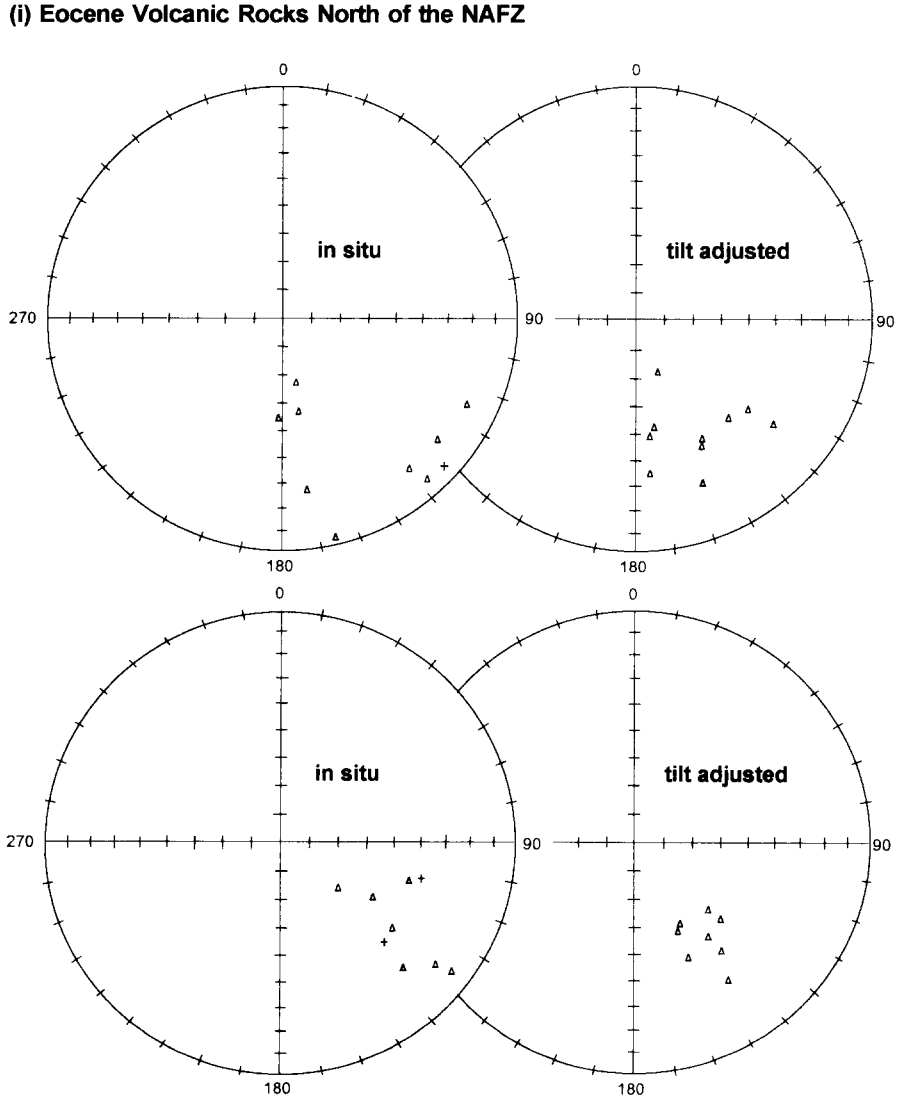


Fig. 5. Site mean directions of magnetisation from Eocene lavas north and south of the North Anatolian Fault Zone (NAFZ) before and after adjustment for tectonic tilt. Triangles are upper-hemisphere projections and crosses are lower-hemisphere projections. Equal-area projections.

provement in precision parameter, k, with tilt adjustment is significant at the 95% confidence level ($k_2/k_1 = 2.87$, McElhinny 1964) and supports a primary origin for the ChRM's in these lavas. The mean inclination moves from -27 to $-43°$ and close to the predicted reverse Early Tertiary field (Table 1). Remanence declinations become more variable at the more highly deformed site locations close to the NAFZ (Table 2 and Fig. 2) and yield a distribution which is more dispersed in declination than in inclination (Fig. 5); this is a signature of unaccommodated plunge components (MacDonald, 1980) and indicates that rotation about local strike does not fully accommodate later deformation.

The remanence declination diverges by about 40° from the Early Tertiary field in a counterclockwise sense and records the summation of later rotations. The only previous palaeomagnetic data from this region were reported by Van der Voo (1968); at that time the volcanic rocks were assigned a Cretaceous age (Geological Map of Turkey, 1:500,000, sheet Samsun, Göksu et al., 1974) and interpreted in the context of a palaeomagnetic field direction of that age. Alternating field demagnetisation of nine samples from two sites yielded in-situ components ($D/I = 166/-44°$ and $D/I = 120/-24°$) broadly comparable with results of this study (cf. Table 2). The direct comparison with the Afro–Arabian Mesozoic data is no longer tenable. Instead a component of regional rotation is evident with respect to both Eurasia and Afro–Arabia (Table 1 and Section 4).

The net regional anticlockwise rotation inferred from these results is the resultant of two sets of motions incorporating (1) the (Miocene) collisional events responsible for emplacement of the Pontides and (2) subsequent anticlockwise rotation of the Anatolian block with respect to the Arabian and Eurasian blocks; the latter motion was first documented in rigid-plate terms by McKenzie (1970) and has been subsequently evaluated in the context of distributed crustal deformation by Rotstein (1984) and others. Analysis of this rotation is considered in Sect. 4.

3.3. Eocene volcanic rocks south of the NAFZ

The lavas comprising the succession at Almus (sites 50–56, 61 and 62) have two component remanence structures in magnetite (Fig. 6) and

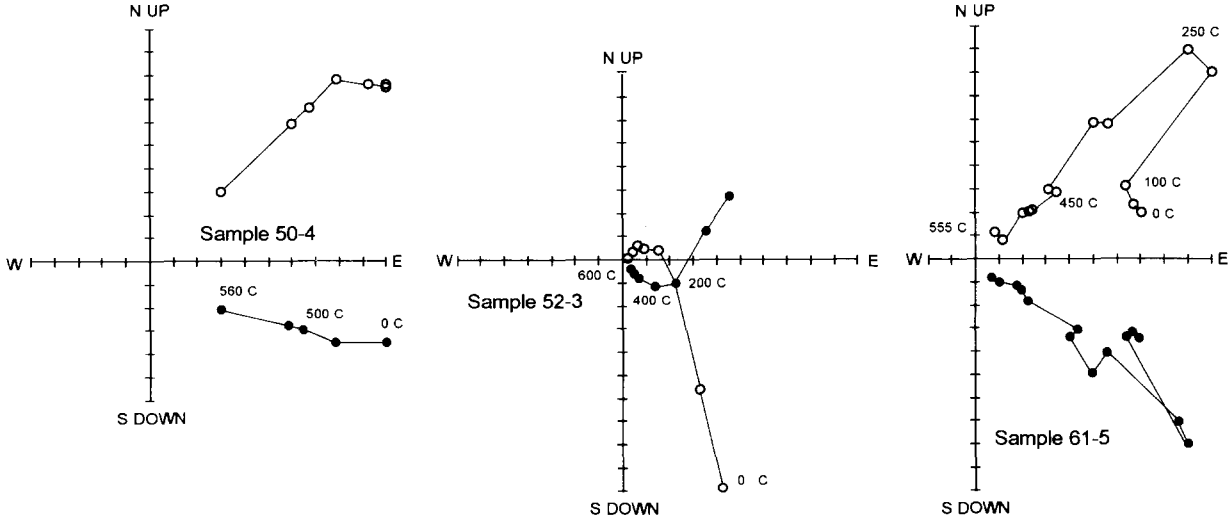


Fig. 6. Orthogonal projections illustrating behaviour of samples from basaltic lavas from the Almus succession south of the NAFZ with progressive thermal demagnetisation. Symbols are as for Fig. 3. Directions are in situ and total NRM intensities are 38.5 (50), 18.6 (52) and 22.0 (61) $\times 10^{-5}$ A m^{-2} kg^{-1}.

convergent high blocking temperature components are largely of reversed polarity (Fig. 5 and Table 2). The highest lava (62) is separated from lower flows by a thin redbed horizon and the remanence moves into a normal quadrant; a partial reversal of the ancient field is implied by weak remanences recorded at this site (2 to 6×10^{-5} A m^{-2} kg^{-1} compared with typical values of 20 to 75×10^{-5} A m^{-2} kg^{-1} at the other sites), suggesting that a transitional field (possibly initi-

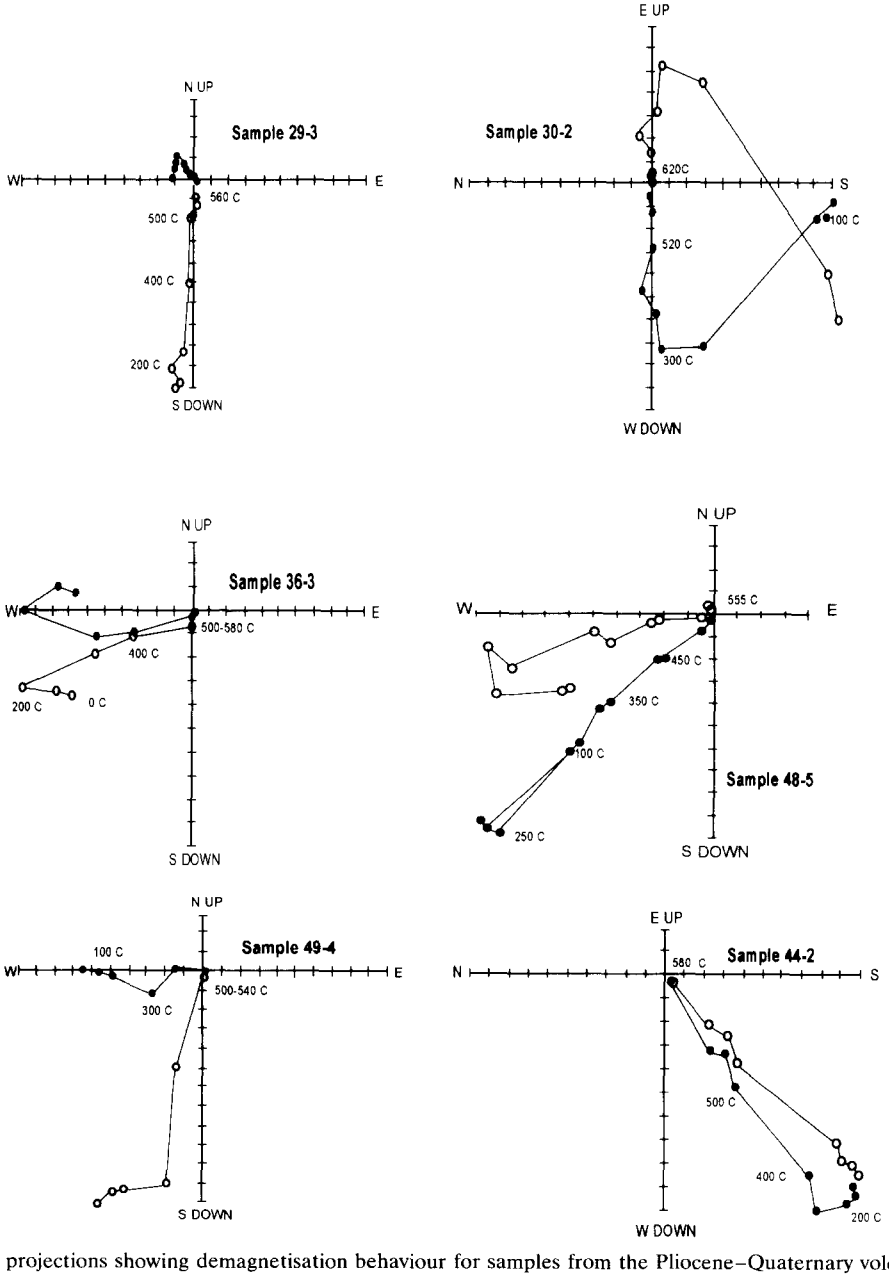


Fig. 7. Orthogonal projections showing demagnetisation behaviour for samples from the Pliocene–Quaternary volcanic rocks from the Niksar district. Symbols are as for Fig. 3 and directions are in situ. Total NRM intensities are 54 (29), 13 (30), 206 (36), 112 (49) and 86 (44) $\times 10^{-5}$ A m^{-2} kg^{-1}.

ating with the shallower inclination at site 61) is recorded here.

The grouping of the remaining site mean directions improves with adjustment for tilt ($k_2/k_1 = 2.19$); because tilts of these lavas are all directed towards the east at intermediate angles, the improvement is not quite significant at the 95% confidence level. The tilt-adjusted inclination moves from -34 to $-48°$ and closer to the predicted Early Tertiary field. The remanence declination is rotated anticlockwise from the Early Tertiary field in common with remanence in the Eocene lavas from the north of the NAFZ (Tables 1 and 2). The amount of rotation, however, is about 8° larger. Although within the α_{95} limits, the difference probably has a tectonic explanation examined in Section 4.1.

3.4. *Pliocene–Quaternary volcanic rocks in the NAFZ*

Eight localities were sampled from the Pliocene–Quaternary volcanic rocks which crop out within the main fault zone and bordering the pull-apart basin (Fig. 2). One site was sampled at Niksar quarry (site 29), one at Hüseyingazi quarry (site 30), four in the Kartankaya hills (sites 43, 44, 48 and 49) and two at Gülbayır village (sites 57 and 58).

The outcrop in Niksar quarry (site 29) is bounded by a pair of strike-slip faults and appears to be located in a transtensional zone between them. Outcrops in the Kartankaya hills (sites 43, 44, 48 and 49) southeast of Niksar town are bounded on the northern side by a NW–SE-trending primary fault. Lavas tilts identified by flow features point towards the southeast, indicating flow onto a young valley floor although tectonic components of dip cannot be resolved precisely. The outcrop in Hüseyingazi quarry (site 30) is located close to Kartankaya and the outcrop at Gülbayır village (sites 57 and 58) is located within the Çanakcı river valley in a feature interpreted as transtensional (Tatar et al., 1990).

Site 29 has a ChRM (Fig. 7) which is not relevant to the palaeofield either in the in-situ orientation or corrected for a dip indicated by flow structures of 35–40° to the south (Table 2). However, if it is assumed that the flow is overturned within this fault-bounded block, the tilt-adjusted direction is moved close to the reversed palaeofield. Hence we infer that this unit, which is the most deformed of the Plio–Quaternary units and is located here against a major fault break with an east–west strike, may have been overturned and possibly rotated clockwise by a small amount. This motion would have taken place at least since the Matuyama reversed chron.

The four sites from the Kartankaya hills (43, 44, 48 and 49) all have W–WSW characteristic remanences of positive inclination (Fig. 7). Although poor exposure precludes precise definition of primary dip, this uncertainty has little effect on the adjusted declinations of these sites (Table 2) and hence would have little effect on the rotations deduced from them. However, inclinations are mostly shallow and hence not fully diagnostic of polarity. We suspect that they are all of normal polarity because the inclinations remain positive even when fully tilt-adjusted; these corrections are probably too large because it is likely that the lavas flowed down primary slopes and there seems to be no justification (Table 2) for arguing that they are of opposite reverse polarity. Hence large rotations are identified from the reference field of normal polarity (Table 2).

The tilt-adjusted result from the nearby lava in the Hüseyingazi quarry (site 30) also shows a similar large anticlockwise rotation of 91°. Sites 57 and 58 lie farther to the northeast along strike from the Niksar site (29). Site 58 yields a normal polarity remanence rotated by a similar amount to sites 30, 43, 44, 48 and 49. Site 57 has probably been overturned and tilt adjustment is difficult to apply; it may possess an overprint in a Recent (normal) field (see in-situ data in Table 2).

Recognition of comparable declinations and probable normal polarity at all but one (the most strongly deformed) of the lavas associated with this young pull-apart basin suggest that igneous activity here has been largely confined to the Bruhnes chron (< 0.78 m.y.) and accords with the lack of evidence for burial. The large rotations within the NAFZ recognised from these rocks

would therefore have occurred during the same period.

4. Discussion and interpretation of the palaeomagnetic results

Palaeomagnetic studies of intracontinental transform faults in western North America (e.g., Luyendyk et al., 1980; Hornafius et al., 1986) and the Middle East (e.g., Ron et al., 1984) have identified distributed zones of large tectonic rotation about the main fault lineaments. Theoretical analysis of deformation in these zones has modelled the lithosphere in terms of a viscous sheet obeying power-law behaviour (England et al., 1985; Sonder et al., 1986). It predicts that rotational deformation should be observed across a broad zone about the transform which is a function of the length of the boundary. In the case of dextral motion across the long NAFZ transform (approximately 1200 km) we expect to observe a broad zone of *clockwise* deformation across the lineament; the width of this zone will depend on the rheology of the lithosphere and the strain rate (Sonder et al., 1986).

4.1. Eocene volcanic rocks bordering the fault zone

In contrast to the predictions of the theoretical models, the formations of Eocene age on both sides of the fault zone show considerable amounts of *counterclockwise* rotation with respect to both the Eurasian and Afro–Arabian plates. These rotations cannot be explained in terms of conventional models of deformation across a dextral transform. It also does not appear to be possible to explain them in terms of back rotation of an earlier and larger anticlockwise rotation because the values of rotation are similar on each side of the lineament (Table 2). The pattern of anticlockwise rotation is suggested by sparse earlier results which, however, are mostly derived from single sites and may not be representative of the time-averaged geomagnetic field (Van der Voo, 1968; Sarıbudak, 1989). The latter author notes a distribution of declinations in the central Pontides north of the NAFZ compatible with the Eurasian plate; declinations appear to swing around in a counterclockwise sense further east along the Pontides and into the region of this study. Hence the counterclockwise rotations observed in the Eocene rocks cannot be attributed to Miocene and younger deformation along the NAFZ. They are therefore likely to be an inherited feature dating from emplacement of the Pontides in pre-Middle Miocene times, although a component of rotation related to back arc spreading in the Black Sea could also be present.

The specific difference between Eocene magnetisations north and south of the fault zone is within limits of palaeomagnetic resolution. It is nevertheless explicable in terms of anticlockwise rotations of the Anatolian block relative to Europe as predicted by the westward escape mechanism of Anatolia described by Şengör and Barka (1992) and summarised in Fig. 8B.

Strike-slip fault systems related to continental escape occur in pairs, forming conjugate escape routes along which continental blocks are expelled sideways and escape-related strike-slip faults commonly end either in extensional areas (e.g., the Aegean to the west of Turkey) or in compressional regions (e.g., the Qilian Shan/Qinghai Nan Shan to the west of Altyn Tagh). Şengör and Barka (1992) have studied the escape systems in Turkey and in Tibet. In both places, the escape-guiding faults (North Anatolian in Turkey, Altyn Tagh in Tibet) display a multibranch "fishbone" structure and both "end" near the terminus of the escape in horsetail structures. The authors note two kinds of fishbone structures, with fundamentally different implications for the architectural evolution of the escaping block.

(1) A *"wedge-in-wedge"* fishbone structure is formed by migration of one of the block-bounding faults into the escaping block. This results in elongation of the original escaping block in the direction of the escape and leads to increased offset along the principal escape fault in the same direction and to incremental decrease in the length of its purely strike-slip part (Fig. 8A).

(2) A fishbone structure created by *rotation of en-echelon wedges* away from the principal fault commonly forms if the escaping block is short-

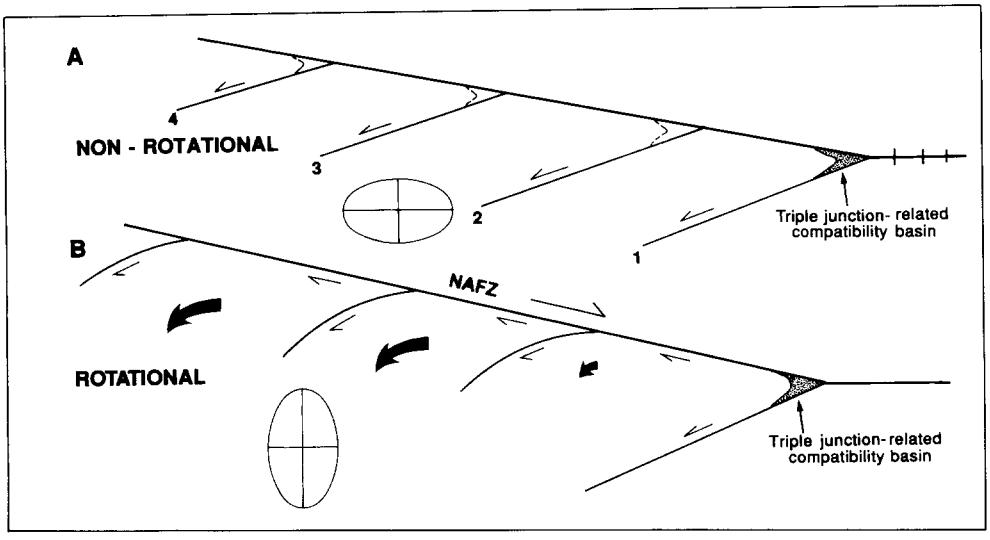


Fig. 8. (a) A "wedge in wedge" kind of fishbone structure. (b) A fishbone structure created by rotation of en-echelon wedges away from the principal fault (Şengör and Barka, 1992).

ened along its escape direction. This leads to a decrease in offset along the principal escape faults and to a shortening in the length of its purely strike-slip segment (Fig. 8B). In both cases basins are formed where fishbone structures intersect the main strike-slip fault.

The second kind of fishbone structure may therefore be applied to explain differential anti-

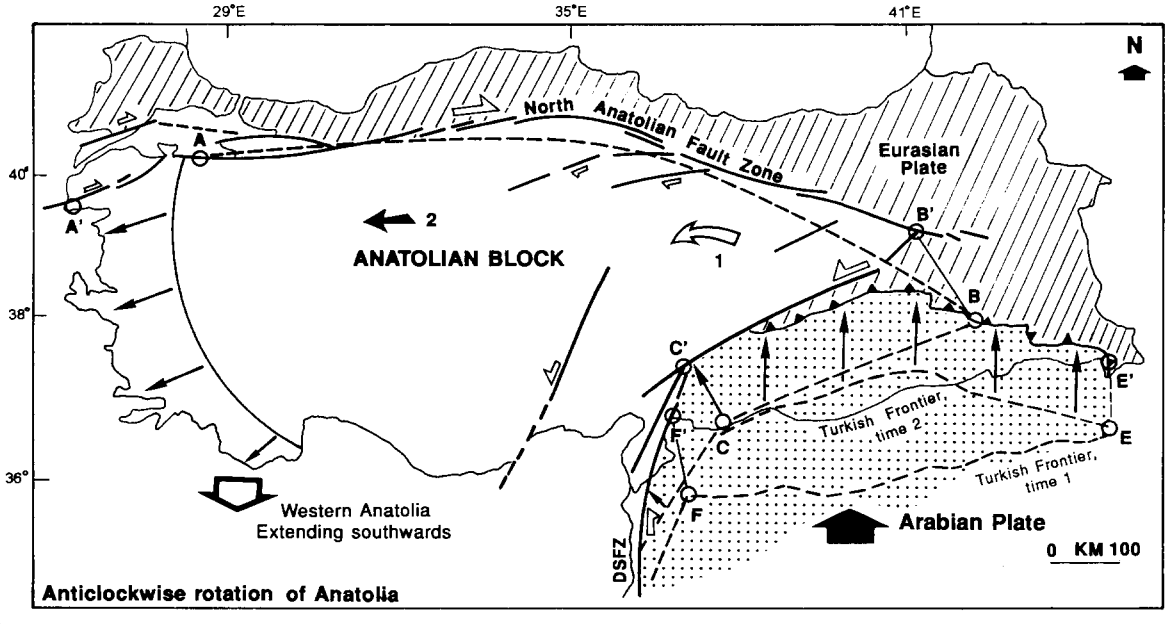


Fig. 9. Possible model to explain anticlockwise rotations exhibited by the Eocene rocks. For simplicity, the motion is visualised in two stages, time *1* and time *2* (present). Northward motion of the Arabian plate (*E–E'*, *F–F'*) causes anticlockwise motion of Anatolia relative to Eurasia (*B–B'*, *C–C'* and *A–A'*).

clockwise rotation of the Eocene volcanics in the present example because the present configuration of strike-slip faults to the west of the Resadiye region (Figs. 1 and 2) is consistent with Şengör and Barka's (1992) second type of fish-bone structure.

In this model anticlockwise rotation takes place involving successively larger blocks with one boundary located respectively at the Almus, Kirikkale–Erbaa, and Taşova–Tekke Fault Zones (all dextral), the other in each case being the NAFZ, which is visualised as extending progressively northwestwards in time. Fig. 9 summarizes how this mechanism may be related to the convergence between the Eurasian and Arabian plates and the westwards escape of Anatolia. Early motion between the plates would have possessed smaller radii of curvature involving displacement on these boundary faults varying from NW–SE to ENE–WSW.

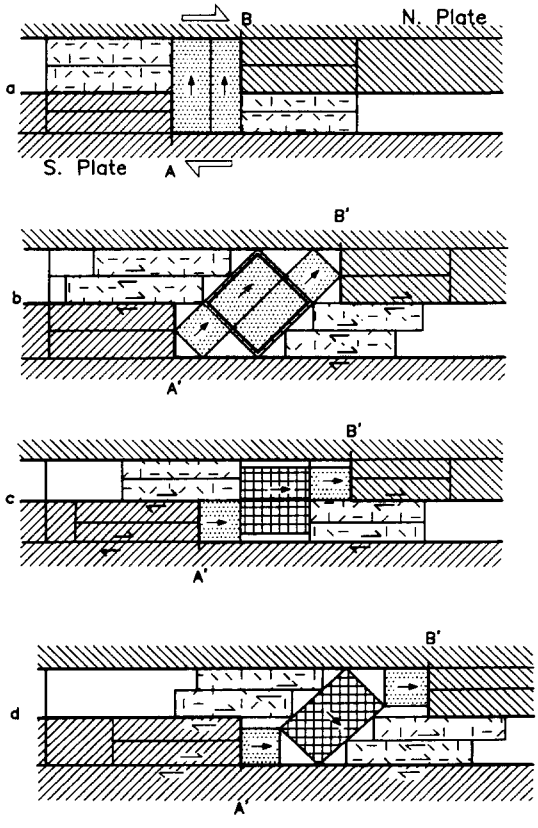


Fig. 10. A model for the large clockwise rotations observed in the Plio–Quaternary volcanic rocks and Cretaceous red limestones within a narrow zone about the NAFZ lineament. The same model can be applied to comparable large clockwise rotations observed in the western sector of the NAFZ (Saribudak et al., 1990).

4.2. Rotations within the NAFZ: Upper Cretaceous and Pliocene–Quaternary units in the fault zone

Upper Cretaceous red limestones. In contrast to the Eocene rocks sampled outside of the NAFZ, these rocks give data interpretable in terms of clockwise rotations of about 100°. Should an earlier anticlockwise phase to the rotation be added (see above), the figure would be increased. The fact that the apparent rotations in each case are clockwise, and presumably post-date Miocene collision, makes it likely that they result from clockwise block rotation induced by dextral motions on the NAFZ.

Pliocene–Quaternary volcanic rocks. Because of the very young age of these volcanics, the rotations inferred from this group must relate to movements on the NAFZ. Large observed variations might be expected if the sites lie in different blocks and the Niksar region (site 29) may well have only experienced a small rotation (+19°), lying as it does at the northern margin of the fault zone. However, sites 30, 43, 44, 48 and 49 (in the Kartankaya hills to the south and east) evidently all lie within a separate block from site 29, and the identified rotations are all remarkably similar ranging from 97 to 124° anticlockwise or from 241 to 263° clockwise.

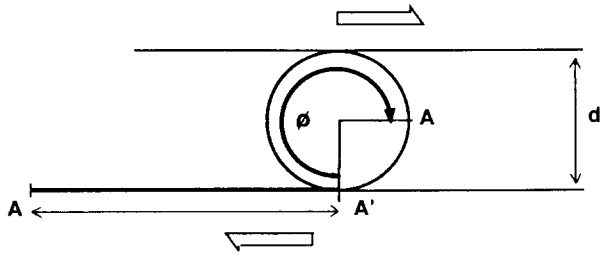
However, there is no obvious mechanism to explain such large anticlockwise rotations because progressive dextral motions within the NAFZ since Miocene times require a clockwise solution. The simplest interpretation of the Pliocene–Quaternary data is therefore that at least two different blocks are involved: an outer block including Niksar, which has rotated by around 19° clockwise, and an inner, central block including the Kartankaya hills which has rotated by about 250° clockwise. A comparable 220° clockwise rotation is observed in the Almacik flake within the NAFZ in western Turkey (Saribudak et al., 1990). The Cretaceous lime-

stones lie in a separate block from the Kartankaya sites and somewhat smaller clockwise rotations are therefore apparent in blocks farther to the southeast.

Large-magnitude block rotations within major strike-slip fault zones have been described from several areas (Şengör et al., 1985; Sarıbudak, 1989; Sarıbudak et al., 1990). Although the possibility of tectonic rotations in excess of 45° have been disputed on theoretical grounds by McKenzie and Jackson (1983) and Ron et al. (1984), they are identified in both western (Sarıbudak et al., 1990) and eastern (this study) segments of the NAFZ. Rotations of up to 45° can be envisaged by progressive clockwise rotation of extensional faults until they achieve an orientation at which point they can act as antithetic sinistral faults (Figs. 10a and 10b). This mechanism requires simultaneous sliding on synthetic NW–SE dextral faults and sinistral movement on rotating antithetic faults. After about 45° the structure locks up, but additional rotation can take place if new extensional faults form and rotate clockwise as before (Figs. 10b and 10c). Thus by the stage represented in Fig. 10d, two blocks have undergone a 90° rotation, while the central large block has experienced rotation of 135°. As the process continues, progressive abrasion of the corners of the block will tend to produce more rounded equi-dimensional blocks which would rotate more freely. The 135° rotation of Fig. 10d represents a total strike-slip displacement of 1.5 times the width of the overlap zone.

The relationship between fault zone width and rotation of a freely moving circular block is shown in Fig. 11. For a block of 5 km in diameter rotating through 270° (cf. the Kartankaya block), the strike-slip displacement would be about 12 km implying average slip rates in excess of 1.6 cm/year if these rocks all belong to the present (Brunhes) polarity chron. Adding a further 5 km for the Cretaceous block gives a total minimum displacement for the Niksar overlap region of about 17 km.

The results of this study show that the NAFZ does not obey a theoretical model for crustal deformation across intracontinental transforms. Instead the predicted rotation is confined to a narrow belt within the main fault zone where rotations much larger than those predicted by theory are observed (McKenzie and Jackson, 1983; Sonder et al., 1986). The absence of distributed rotation here contrasts with the distributed deformation observed along some other transform plate boundaries (Ron et al., 1984; Hornafius et al., 1986; Kissel and Laj, 1988). As a consequence of the mechanical behaviour of the lithosphere this may reflect the asymmetry of the other studied examples where only a narrow continental block is present on one side; the NAFZ in Turkey is bounded on both sides by wide continental blocks. The concentration of deformation within a narrow zone about the NAFZ appears to be corroborated by the evidence from historic earthquake activity which is accommodated by movements on sub-parallel fault planes within a narrow zone about the main lineament (Jackson and McKenzie, 1984).



Displacement AA' = $\emptyset/\pi$ d , where $\emptyset$ is the angle of rotation expressed as a proportion of 360°and d is the diameter of the block.

Fig. 11. The relationship between fault zone width and rotation of a freely moving circular block (the $L\phi = 270°$).

Acknowledgements

We are grateful to Mrs Joan Dean of the Geomagnetism Laboratory, University of Liverpool, for undertaking some of the laboratory work for this study. Prof. Dr. A.M.C. Şengör (ITÜ) kindly provided some of his unpublished figures and we thank Dr. C. Kissel and Dr. C.W.T. Klootwijk for helpful reviews of the manuscript.

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